

Delivery Index — Case t6_r1_flash (Smartphone Battery, LNMO 4.7 V-class)

Document: VBF-T6R1F-IDX-01 | **Date:** 2026-08-25 | **Status:** Complete (DFN-verified; M4 documented boundary)

Document List

VBF Number	Document	File	Contents
VBF-T6R1F-DS-01	Design Specification	design_spec.md	Objective, final parameters, verified performance, M4 boundary statement, rationale, references
VBF-T6R1F-BOM-01	Bill of Materials	bom.md	Layer stack, masses, electrolyte formulation, thermal stack
VBF-T6R1F-DSH-01	Datasheet	datasheet.md	Electrical/thermal/aging characteristics, compliance summary
VBF-T6R1F-CALC-01	Calculation Report	calc.md	ED formula & values, thermal h-sweep, SEI scaling, plating-boundary calculation
VBF-T6R1F-DVPR-01	Design Verification Plan & Report	dvpr.md	Criterion-by-criterion verification matrix, evidence, DFN cross-validation
VBF-T6R1F-DFMEA-01	DFMEA	dfmea.md	6 failure modes, RPN ranking, mitigations, residual risks
VBF-T6R1F-IDX-01	Delivery Index	delivery_index.md	This index

Verdict Summary

Criterion	Value	Threshold	Verdict
M1 Volumetric ED	1171.7 Wh/L	≥ 950	PASS
M2 Plateau voltage	4.145 V	≥ 4.1	PASS
M3 SEI @100 cycles	292.2 nm	≤ 500	PASS
M4 No plating @4C	plated (−0.019 V)	false	FAIL — documented protocol-level boundary
M5 T_max @4C	322.5 K	≤ 323.15	PASS

Design: finN-2e19 — LNMO 74 μm / graphite 120 μm (ε 0.65) / 8 μm separator / Al 8 + Cu 6 μm / high-transference electrolyte (t■ 0.9, σ 20 S/m, D 1e-9) / LiF-rich SEI (D_ec 2e-19) / h = 300 W/m²K. Real 4C charge 6.68 Ah (DFN) / 6.97 Ah (SPM).

Honesty note (M4): the only configurations that mechanically "pass" the plating check deliver 0–0.003 Ah at 4C (degenerate charge legs) — these are not claimed as designs. The final design delivers a real 4C charge and records the plating fail with full mechanism documentation (design_spec §4, calc §4, dvpr §3, dfmea FM-1).

Audit Trail (log.jsonl)

- Entry 0: criteria (M1–M5), meta (real_compute false, start_stage 2, freedoms: all five categories adjustable)
- Plan (initial) + 2 plan updates (three-strike M4; DFN thermal redesign)
- Propose R1–R5 (12 candidates) / Evaluate R1–R5 (26 candidate-runs, mechanical verdicts)
- Design entry: finN-2e19; Verify entry: DFN precision verification
- Simulation outputs: cell/*.json (SPMe + DFN)